Richard Vidal-Dorsch

October, 2026 · @Richard Vidal-Dorsch

PRINCIPAL EMBEDDED & RUST SOFTWARE ENGINEER

964 Springfield St., Costa Mesa, CA 92626 · (714) 829-6831 · richard.dorsch@gmail.com · LinkedIn · rvido.github.io · github.com/rvido

Professional Profile

Principal-level embedded software engineer with 20+ years of experience designing and optimizing resource-constrained, performance-critical systems. Deep expertise in production Rust and C/C++, embedded Linux, real-time operating systems (RTAI, VxWorks, QNX), Linux kernel development, and network security. Combines hands-on implementation with system architecture, technical leadership, and build/CI infrastructure ownership.

- **Performance engineering:** Zero-allocation packet processing, lock-free data structures, and measured throughput exceeding 1.6 million packets/s on a quad-core ARM Cortex-A72.
- **Reliability and verification:** 2,000+ automated tests, seeded fuzzing, and independent validation of kernel-executed eBPF programs.
- **End-to-end systems engineering:** From bootloaders, BSPs, and kernel drivers to optimized Rust applications, cross-compilation, deployment, and on-target profiling.

Technical Skills

- **Languages:** Rust (`no_std`, async Rust/Tokio), C (C99), C++ (C++11/17), Python, Go, Bash/Zsh, Assembly (ARM64, 68k, x86), Objective-C++
- **AI-Assisted Engineering:** AI-assisted coding, refactoring, debugging, test generation, code review, technical documentation, and development workflow automation
- **Embedded Systems:** Embedded Linux, Linux kernel drivers, BSPs, U-Boot, bare-metal AArch64, RTAI, VxWorks, QNX, OpenWrt
- **Architecture & Performance:** Real-time systems, zero-allocation and zero-copy design, lock-free data structures, on-target profiling, benchmarking, flamegraphs, link-time optimization (LTO), static linking, musl
- **Networking & Security:** TCP/IP, raw sockets, libpcap, PACKET_MMAP, eBPF, nftables, NFQueue, DNS, TLS fingerprinting (JA4/JA4S), QUIC, WireGuard, OpenVPN-3, Ed25519
- **Hardware & Interfaces:** ARM Cortex-A/M, Raspberry Pi CM4/CM5, x86/x86-64, MIPS32, PPC/VME, TI C67xx DSP, ESP32; I²C, SPI, GPIO, UART, RS-232, USB
- **Build Systems & Automation:** Cargo, CMake, Make, cross-compilation, Docker, Jenkins, GitHub Actions, Git, CI/CD, Prometheus
- **Testing & Analysis:** Criterion, fuzz testing, AddressSanitizer, automated integration testing
- **Development Tools:** JIRA, Confluence
- **Languages:** English, German, French, Spanish

Professional Experience

AncileNet | Costa Mesa, CA — **Founder & Principal Software Engineer** | March 2026–Present

Designed and implemented AncileNet, a ~56,000-line Rust network security engine for embedded gateways, deployed as an ~8.1 MB statically linked AArch64 binary on a Raspberry Pi CM4 running OpenWrt.

- **Throughput and efficiency:** Engineered a zero-allocation packet-processing path using borrowed slices over capture ring buffers, achieving 1.61 million packets/s across four Cortex-A72 cores, with 80% parallel-scaling efficiency.
- **Kernel-level optimization:** Reduced engine CPU consumption by 26× (20.7% to 0.8% of one core) with a 186-instruction eBPF socket filter that rejects 99.3% of irrelevant packets before userspace processing. The JIT-compiled program occupies 1,168 bytes.
- **On-target profiling:** Built a sampling profiler because the target kernel lacked `perf`; identified device discovery as a major bottleneck and reduced its CPU share from 38% to 6.5%.
- **Capture reliability:** Eliminated packet-capture losses of approximately 33% under load by sizing kernel buffers according to available system memory and adding per-cause drop telemetry.
- **Verification:** Maintained 2,045 passing unit, integration, and seeded-fuzz tests, built a test-only eBPF interpreter to validate the shipped kernel program against 20,000 generated and corrupted frames, catching a length-calculation bug.
- **Network security:** Implemented JA4/JA4S TLS fingerprinting, TCP handshake reassembly, QUIC Initial decryption (RFC 9001), Ed25519-signed update manifests, and nftables kernel-set enforcement.
- **Security-conscious architecture:** Rejected load-driven buffer resizing after identifying an attacker-triggerable inspection gap, prioritizing inspection integrity over throughput under adverse conditions. Delivered a REST API and web dashboard for operational visibility.
- **Architecture:** Defined system architecture across firmware, networking, and application layers.
- **Engineering practices:** Established standards for testing, code quality, and cross-platform builds.
- **Hardware/software integration:** Coordinated integration across hardware, firmware, and application components.
- **AI-augmented development:** Leveraged AI-assisted coding, debugging, refactoring, and test development to accelerate a production-oriented Rust network detection engine in an eight-month development cycle, while retaining responsibility for architecture, implementation decisions, and validation.

Aura Company | Burlington, MA — **Senior Software Engineer** | 2021 – Feb 2026

- Led architectural decisions across the firmware, cloud, and application layers.
- Mentored engineers in embedded development, debugging, performance analysis, and build-system design; supported modernizing legacy components in Rust to improve memory safety and maintainability.
- Optimized Objective-C++ VPN tunnels for iOS and Android using Apple's Network Extension framework, maintaining agent memory usage below 30 MB while supporting high-throughput packet parsing.
- Interfaced C/C++ core libraries with Swift/Objective-C via bridging headers and Objective-C++ wrappers.
- Architected a Rust Platform Builder to scan and categorize ranked network domains for mobile app integration.
- Engineered C lookup systems with dynamically scaled vectors and hash tables for real-time parental controls and fraud detection.
- Designed a Docker-based Firmware Builder and Domain Database integrated with GitHub Actions CI/CD.
- Led the re-implementation of the Domain Categorizer in Rust for memory safety and maintainability.

Circle Media Labs Inc. | Cypress, CA — **Senior Firmware Engineer** | 2017 – 2021

- Led next-generation hardware design and ported parental-control agents to Netgear router platforms.
- Coordinated hardware/software integration and resolved issues between application development and embedded teams.
- Moved builds to CMake with a configurable package builder, reducing cross-platform build complexity by 80%.
- Configured cross-compilation toolchains for embedded targets and mobile frameworks.
- Built a Docker-based Jenkins build server supporting T-Mobile, Sky Broadband, and Netgear.
- Reduced domain database build times by 90% through Python optimization; mentored engineers on AWS integration.

Advantech Corp. | Irvine, CA — **Senior Software Engineer (Embedded Linux)** | 2007 – 2017

- Led embedded Linux projects for Tier-1 clients, including Seagate, GE, Amazon, Facebook, and Cisco.
- Wrote and upstreamed the iManager Linux kernel driver suite; developed I2C, GPIO, and watchdog drivers and Android/U-Boot bootloaders.
- Established cross-platform toolchains for i.MX31/53, the foundation for driver development across the product line.

NDC Infrared Engineering | Irwindale, CA — **Senior Embedded Software Engineer** | 2007

- Developed high-precision X-ray measurement software for industrial rubber thickness on TI C67xx DSPs.

KIMTEC (contract for Panasonic Avionics Corp.) | Lake Forest, CA — **Application Software Engineer** | 2006

- Developed core communication servers for in-flight entertainment systems, enabling aircraft-to-ground connectivity.

J. Müller AG | Effretikon, Switzerland — **Software Engineer / System Administrator** | 2002 – 2006

- Delivered real-time Linux (RTAI) solutions and system/database administration across cross-functional projects.
- Programmed low-level components in 68k/x86 assembly on VME-bus/PPC systems for audio/video processing.

Open Source Projects

- **LANwatch (Rust):** Network discovery engine decoding 18 link-layer and discovery protocols directly from raw frames, with no unsafe blocks in the project's Rust source. Features bounds-checked parsers, protocol-specific Cargo features, and size-optimized builds; integrated into AncileNet.
- **memtracer (C99/C++11):** Single-header, zero-heap memory diagnostics engine for embedded and safety-critical targets. Static record buffer and O(1) open-addressing hash table, transparent `malloc`/`free`/`new`/`delete` interception with call-site capture, spinlock thread safety, and clean AddressSanitizer runs.
- **Bare-metal Rust on Raspberry Pi 4 (Rust, no_std):** Boot proof-of-concept for BCM2711/Cortex-A72 on stable Rust (`aarch64-unknown-none`) with a custom linker script and PL011 UART driver.
- **iManager (C, Linux kernel):** Core Linux driver suite and hardware abstraction layer for Advantech embedded boards; driver patches upstreamed.
- **BME280_Linux (C++17):** Linux driver for the BME280 environmental sensor over I2C/SPI.
- **OpenWrt firmware updater (Shell):** Automated backup, ext4 image validation, and sysupgrade for field updates of OpenWrt devices.
- **whois-rdap, NetOrigin, AppID_finder (Rust):** CLI and web tools for RDAP/WHOIS lookups, IP ownership and ranges, and iOS/Android bundle-ID discovery.

Education

FHS St. Gallen University of Applied Sciences | St. Gallen, Switzerland

- Executive Master of Science in Software Engineering — software architecture, IT management, enterprise systems.
- Bachelor of Science in Electrical Engineering — embedded systems, digital signal processing, microelectronics.

Awards & Community

- **Employee Excellence Award**, 2009 and 2013 — Advantech's highest recognition, for mission-critical embedded Linux work for Tier-1 clients.
- Upstream Linux kernel contributor; regular participant in the Embedded Linux Conference community.